

Advanced Financial Econometrics

SUMMER INTENSIVE MODULE • 2026



VOLATILITY CLUSTERING • Σ^2_t

29 JUN – 17 JUL 2026

IN-PERSON

MON-FRI

4 UNITS

01 INSTRUCTOR

Mehman Ismayilli

Module Lead • Financial Econometrics

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OFFICE HRS 16:00–17:00 • Mon • Tue • Wed

FORMAT In-person lectures & seminars

02 AT A GLANCE

66

CONTACT HOURS

4 units

CREDITS

21/wk

SELF-STUDY HRS

3 wks

DURATION

03 LEARNING OUTCOMES

- 01 Construct and interpret ARMA and GARCH models to analyze financial returns.
- 02 Forecast market volatility using econometric techniques.
- 03 Model derivative pricing using continuous-time stochastic processes.
- 04 Apply EVT and VaR techniques to assess market risk and tail behavior.
- 05 Communicate empirical findings through written and oral presentation.

04 SCHEDULE & DAILY STRUCTURE

17:10 – 18:40 • Lecture → 10 min break → 18:50 – 19:50 • Seminar =

2h 40m / day

29 JUN

Term begins

TUE • 07 JUL

Midterm Exam

16 JUL

Project Demo + Reflection

17 JUL

Final Exam

05 REQUIRED TEXT

PRIMARY Tsay, R. S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley Series in Probability & Statistics.

ISBN 978-0470414354

06 SYLLABUS BY WEEK

WEEK 1

Financial Time Series and Their Characteristics

Tsay · Ch 1

Linear Time Series Analysis and Its Applications

Tsay · Ch 2

WEEK 2

Conditional Heteroscedastic Models

Tsay · Ch 3

WEEK 3

Nonlinear Models and Their Applications

Tsay · Ch 4

Continuous-Time Models and Their Applications

Tsay · Ch 6

07 ASSESSMENT

COMPONENT	DATE	WEIGHT
Attendance Continuous, all sessions	Ongoing	12%
Midterm Exam In-class, written	Tue 07 Jul	20%
Project Demonstration Applied analysis in R	16 Jul	20%
Course Reflection Written, self-assessed	16 Jul	18%
Final Exam In-class, written	17 Jul	30%
Total		100%

08 SOFTWARE SETUP · R & RSTUDIO

STEP 01 · INSTALL R

R base distribution

Download from CRAN at cran.r-project.org. In mainland China, a local mirror is faster and more reliable.

RECOMMENDED MIRRORS (CHINA)

Tsinghua `mirrors.tuna.tsinghua.edu.cn/CRAN`

USTC `mirrors.ustc.edu.cn/CRAN`

Aliyun `mirrors.aliyun.com/CRAN`

STEP 02 · INSTALL RSTUDIO

RStudio Desktop (free)

Install R first, then download the IDE from Posit at posit.co/download/rstudio-desktop.

SET THE MIRROR INSIDE R

```
options(repos =  
"https://mirrors.tuna.tsinghua  
.edu.cn/CRAN")
```

09 COURSE MATERIALS

Data & R code for in-class exercises

Download the datasets and scripts before each seminar.

mnismayilli.github.io/lectures/FE